Upesh Jeengar

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Summary

Data Scientist at **FICO** with expertise in developing **scalable ML solutions**, **generative AI pipelines**, and **cloud-native applications**. Skilled in bridging complex **mathematical modeling** with **production-grade engineering**, including expertise in **Java**, **Python**, and **AWS** infrastructure. Proven track record of implementing advanced **LLM architectures—**including **RAG** and **SFT—**to solve real-world financial challenges.

Technical Skills

Languages: Python, C/C++, SQL, JavaScript, HTML/CSS, Terraform, Java, Bash/Shell

Developer Tools: Git, GitHub, CI/CD (GitHub Actions), Docker, AWS (IAM, S3, EC2, Lambda, EFS, Bedrock, VPC, SageMaker), Azure, Linux, VS Code, Visual Studio, Claude Code, Opencode

Libraries & Frameworks: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, PyTorch, MLflow, DVC, Transformers, Datasets, Flask, Gunicorn, Nginx, LangChain, LangGraph

AI: LLM Architecture, SFT, GRPO Fine-tuning, Agentic AI, Google ADK, RAG, Vector DBs, MCP, Computer Vision, Unsloth, LLMOps, vLLM, GPU Computing on EC2 and Azure VMs

Other Tools: Excel, Word, Tableau, PowerPoint

Professional Experience

Fair and Isaac Corporation (FICO) — ***July 2025 - Present***

Analytics Science – Associate Data Scientist

AWS-Based Infrastructure Setup for Fraud Detection Tools

- Made a **250+ tool codebase platform-independent** for deployment on **AWS Mumbai EC2** infrastructure, enabling **FICO Falcon Fraud Manager** tooling to comply with **RBI data residency** requirements while supporting dynamic path and parameter resolution based on host environments. Setup **cron jobs**, **configuration scripts**, **AD user migration**, and secure permissions.

Intelligent Loan Approval System Using Predictive Analytics

- Built an **Intelligent Loan Approval System** by integrating a **Logistic Regression** model into a **Spring** application, using **feature engineering**, **TF-IDF**, and **recursive feature elimination** to achieve **77% accuracy**. Leveraged an internal **LLM** to provide **model explainability**.

Pseudocode Generation with Amazon Bedrock

- Built an automation script to recursively resolve dependencies of complex **credit risk variables** (**credit**, **mortgage**, and **bank charge** amounts and ratios) and fed the resolved logic to **Amazon Bedrock**-hosted **LLMs**, automatically generating human-readable **pseudocode documentation** for 400+ financial variables for **Westpac Bank Australia**.

NGMA Synthetic Data Generation

- Developed a configurable **Python**-based **synthetic data generation pipeline** for **NGMA**, producing realistic hierarchical product datasets with spatiotemporal purchase behaviors, seasonality, and **Pareto-driven distributions** for analytics. Created a high-quality product purchase dataset with 100 million records. Optimized the pipeline for **30% faster generation**.

GitHub Actions Migration & CI/CD Modernization

- Led a **GitHub Actions** POC to migrate **FICO AIID** repositories from **Bitbucket** to **GitHub**, projecting an 75% reduction in total cost. Executed the transition by converting **Jenkins CI** pipelines for 4 repositories into **GitHub workflows** integrated with **Jira**, **JFrog**, and **Black Duck/Checkmarx** security scans. Configured organization wide self-hosted **CPU/GPU EC2 runners**.

Education

Indian Institute of Technology (IIT) Guwahati — ***Nov 2021 - July 2025***

- *Bachelor of Technology (B.Tech.) in Mathematics and Computing*

Projects

VLM from scratch | *Python, LLMs* — ***Dec 2025 - Jan 2026***

- Built a **BLIP-2**–style **Vision-Language Model** by contrastively training a **Q-Former** to align **ViT** image embeddings with text.
- Trained the system in two stages: first aligning image and text features, then fine-tuned **SmolLM-135M**.
- Trained and evaluated the model on **50K Conceptual Captions** pairs, achieving **90% Recall@5**.

End-to-End Flight Price Prediction | *Python, Flask, AWS* — ***Oct 2023 - Nov 2023***

- Built a **Random Forest** model for flight price prediction with an **R^2 score of 73.65%**. Used **MLflow** for model monitoring, ensuring robust evaluation and **hyperparameter tuning**. Operationalized a scalable **Flask API** on **AWS EC2**, utilizing **Gunicorn** and **Nginx** for production.